LAB 06 - PROMPTS

Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab06-pine-strategy-tester/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Read the strategy code (A2, K3)

Prompt - copy into your AI coding assistant
Explain labs_fullcode/Lab06-pine-strategy-tester/lcc_ma_crossover_strategy.pine for a finance professional: each strategy() setting, when an order is filled (process_orders_on_close = false means the NEXT bar's open), and how commission and slippage reduce the result. Do not change the file.

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Check: You can state that a signal at Monday's close is filled at Tuesday's open, with USD 1 commission per order.

Step 2: Run it in the Strategy Tester and record the results (A3, K7)

Do this (no prompt needed)
Open AMEX:SPY on 1D, paste the script into a new Pine Editor tab, save and Add to chart. Open the Strategy Tester panel and copy net profit, closed trades, percent profitable, profit factor, max drawdown, Sharpe, Sortino and commission paid into Lab06-pine-strategy-tester/tv_strategy_results.yaml (replace the nulls).

Check: The chart shows SMA 50 / SMA 200, green shading while in a position, and the Performance Summary tab.

Step 3: Build the reconciliation stage (A2, A3, K3)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/reconcile.py. parse_pine(path) reads the //@version, the keyword settings inside strategy(...) and the input.int defaults (fastLen, slowLen) from Lab06-pine-strategy-tester/lcc_ma_crossover_strategy.pine. settings_checks(pine, cfg) returns PASS/FAIL rows: version 6; process_orders_on_close = false (next-bar-open fills = config strategy.execution next_open); commission_type cash_per_order with the value in config costs.tradingview; slippage > 0; default_qty_value 100 (% of equity); pyramiding 0. run(symbol='SPY', fast=None, slow=None) backtests the same rule with common.backtest.backtest(bars, cross_signal(ta_sma(fast), ta_sma(slow)), costs=pipeline.config.tradingview_cost_model(cfg), execution='next_open') and buy_and_hold, prints the Strategy Tester equivalents (net profit USD and %, closed and open trades, percent profitable, profit factor, max drawdown, Sharpe, Sortino, commission, buy and hold) next to the values in Lab06-pine-strategy-tester/tv_strategy_results.yaml with a difference column, the trade list, and four likely causes of gaps (adjusted vs unadjusted prices, 2018 history start, whole vs fractional shares, ticks vs bps slippage). Write _outputs/Lab06_reconciliation.yaml (plain Python numbers) and chart Lab06_crosscheck.png with common.plotting.plot_equity_and_drawdown. Exit code 1 if a settings check fails. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline reconcile --help shows --symbol, --fast and --slow.

Step 4: Reconcile TradingView with the pipeline (A2, A3)

Do this (no prompt needed)
Run: python -m pipeline reconcile --symbol SPY --fast 50 --slow 200 (same as: python3 Lab06-pine-strategy-tester/reconcile_stage.py).

Check: 6 settings checks PASS. Pipeline: net profit USD 79,878 (+79.88%), 4 closed trades + 1 open, 50.0% profitable, profit factor 5.03, max drawdown -33.72%, Sharpe 0.55, Sortino 0.76, commission USD 9.00, buy and hold +187.26%; first entry 2018-10-17; your TradingView figures appear in the TradingView column with differences.

Step 5: Explain the differences (A3)

Prompt - copy into your AI coding assistant
Here is my reconciliation output: <paste the table from python -m pipeline reconcile>. For each metric where TradingView and the pipeline differ by more than 5%, say which of these causes is most likely and how I could confirm it: adjusted vs unadjusted prices, the 2018 data start and SMA 200 warm-up, whole vs fractional shares, slippage in ticks vs bps.

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Check: Each gap is linked to a named cause; typically TradingView's longer history changes the first trade.

Step 6: Stress the costs and the speed of the rule (K7)

Do this (no prompt needed)
In TradingView change commission to 0.1% of order value, slippage to 5 ticks and the inputs to 20/50. Then run: python -m pipeline reconcile --fast 20 --slow 50.

Check: Pipeline SMA 20/50: net profit USD 88,228 (+88.23%), 16 closed trades, max drawdown -27.84%, Sharpe 0.72, commission USD 33 - more trades pay more cost; record how TradingView's net profit falls with 0.1% commission.

Step 7: Put the strategy into action with TradingView Paper Trading (A3)

Prompt - copy into your AI coding assistant
Act as my execution checklist for TradingView Paper Trading on AMEX:SPY with USD 100,000. Using the last pipeline close of 676.64, give the order-ticket fields and prices for (1) a market buy of 10 shares, (2) a limit buy 2% below the last price, (3) a bracket buy with take-profit +5% and stop-loss -3%; then say which status each order should show in the Orders and Positions tabs.

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Check: Market buy filled in Positions; limit buy (about 663.11) Working in Orders; bracket with take-profit about 710.47 and stop-loss about 656.34 shown as lines on the chart; commission deducted in the account summary.

Step 8: Manage the paper position from the chart (A3, K7)

Do this (no prompt needed)
Drag the stop-loss line to move it, close half the position, then close all. Read the History tab and the account balance. Note: a Pine strategy does not send orders to the Trading Panel by itself - automated orders go through alerts and webhooks (Lab 17) or the pipeline's execution stage (Lab 8).

Check: History lists every fill with its commission; the position is flat and the balance reflects P&L and costs.

Step 9: Optional - connect the Alpaca paper account in TradingView (A3)

Do this (no prompt needed)
In the Trading Panel choose Alpaca, log in, select the PAPER account and place one small market or limit order from the chart. Then run python -m pipeline check (online) to see the same position from the pipeline.

Check: The order appears in the Alpaca paper dashboard and as an open position in section [4] of the check.

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Step 10: Decide whether to promote the rule (A2)

Do this (no prompt needed)
Apply the strategy to SGX:D05 and compare. Decide whether SMA 50/200 should replace the pipeline's configured ema_cross 10/30 (do not change config.yaml yet - Labs 9 and 10 measure it properly).

Check: A one-paragraph decision that cites net profit vs buy and hold (+79.88% vs +187.26%) and the drawdown.

Run it

Step	Command / file
Run Lab 06	python3 Lab06-pine-strategy-tester/reconcile_stage.py (same as: python -m pipeline reconcile)
The complete code	pipeline/reconcile.py (reference copy: Lab06-pine-strategy-tester/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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